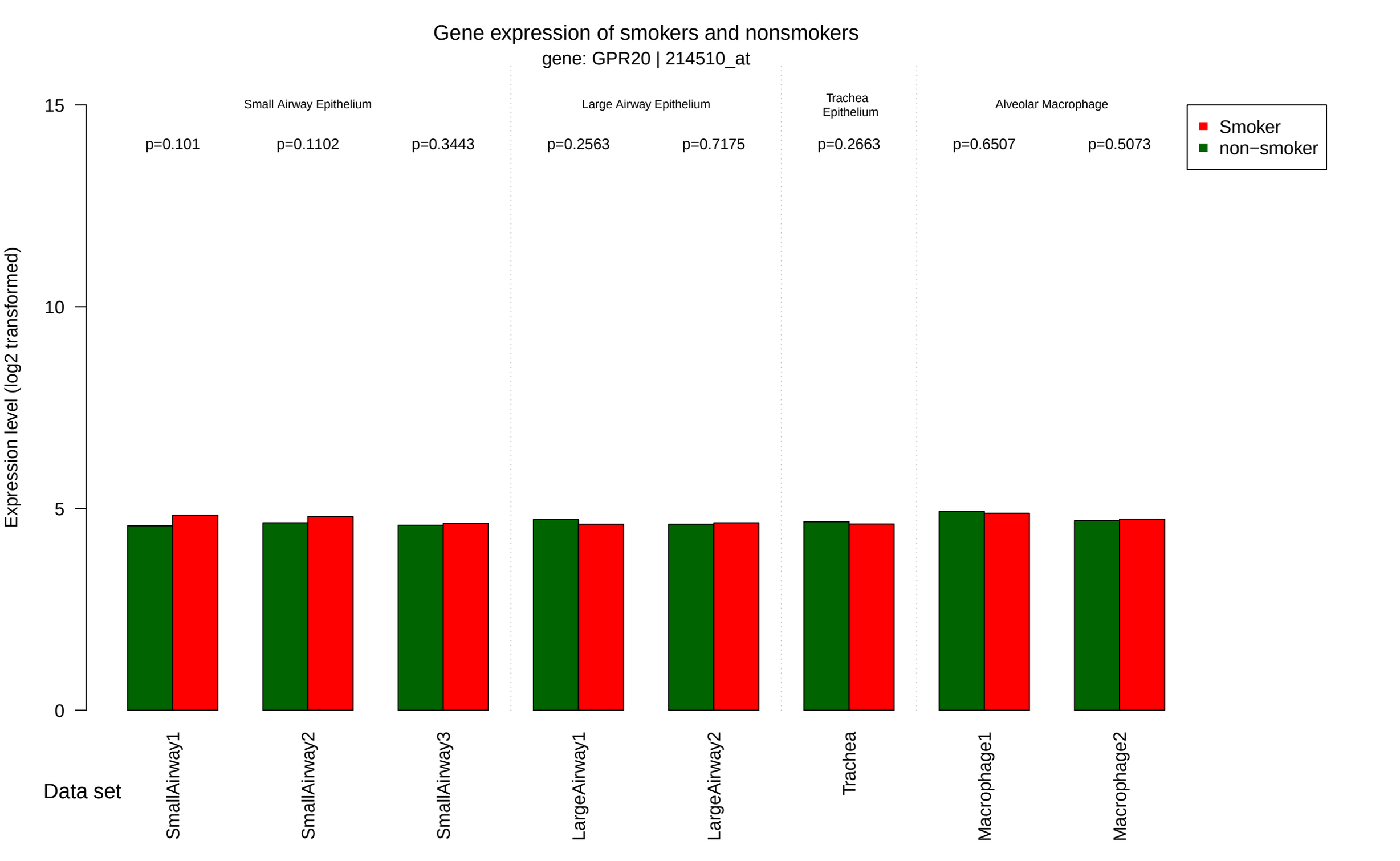
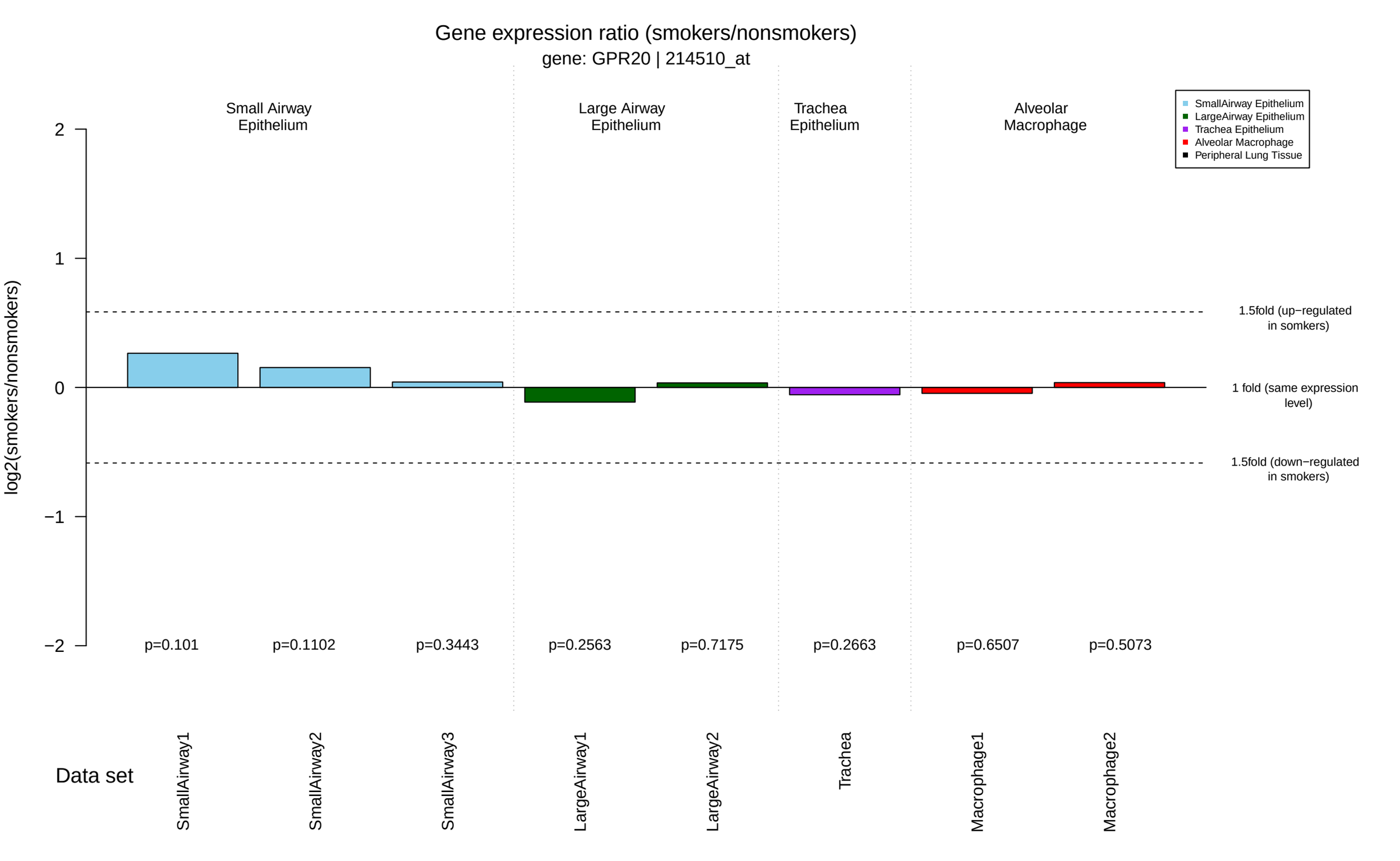
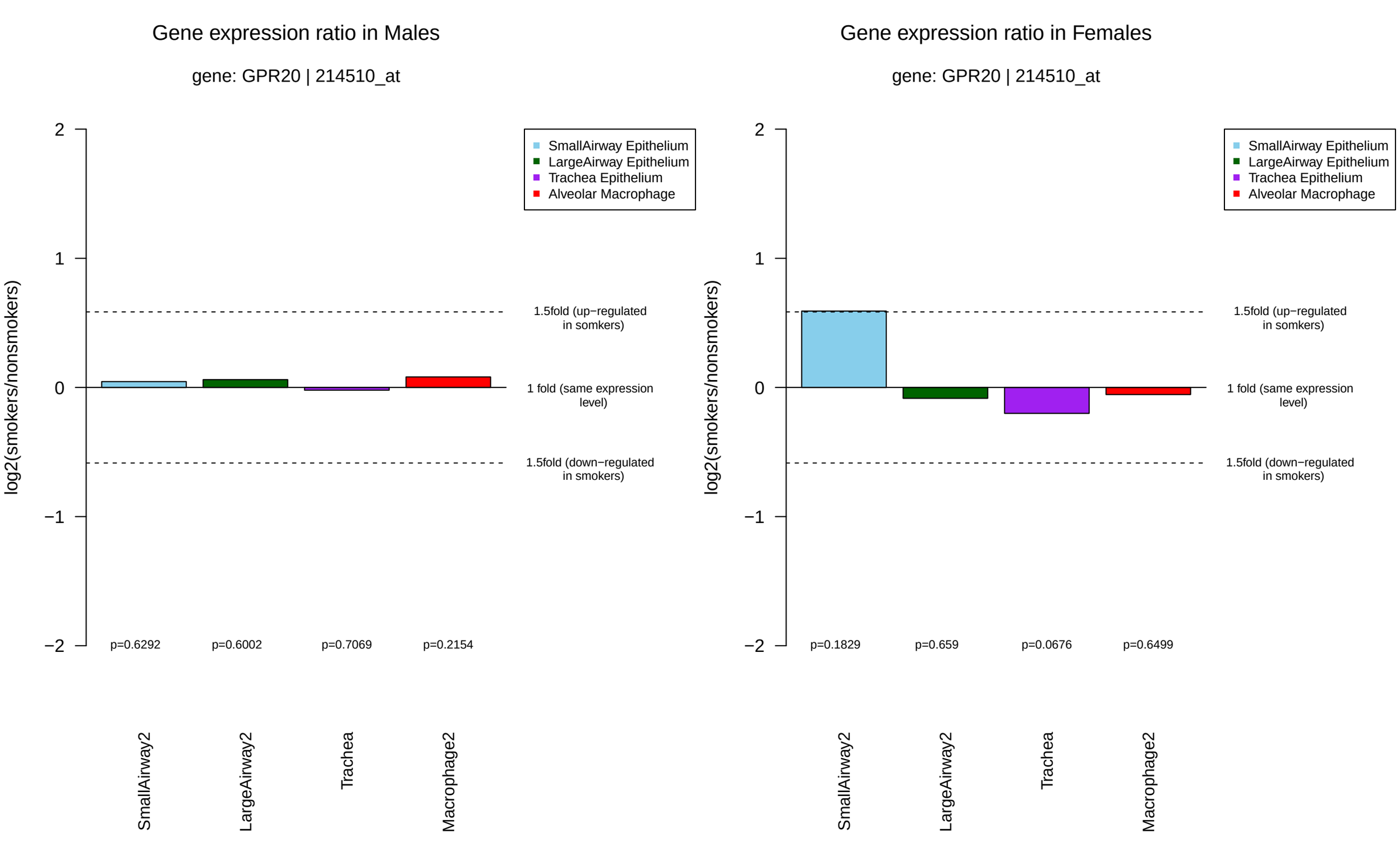
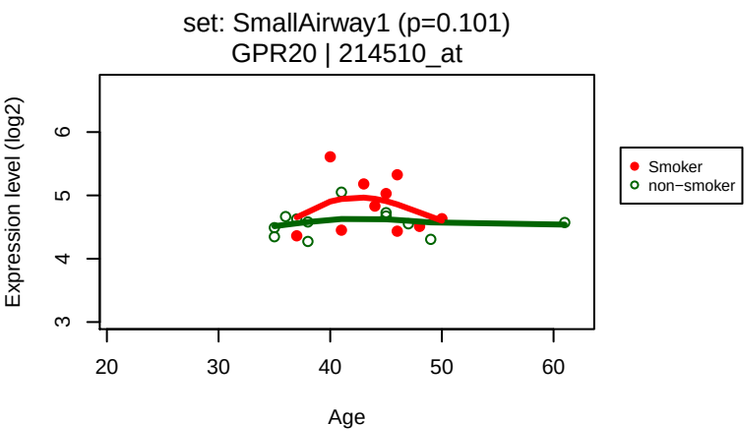
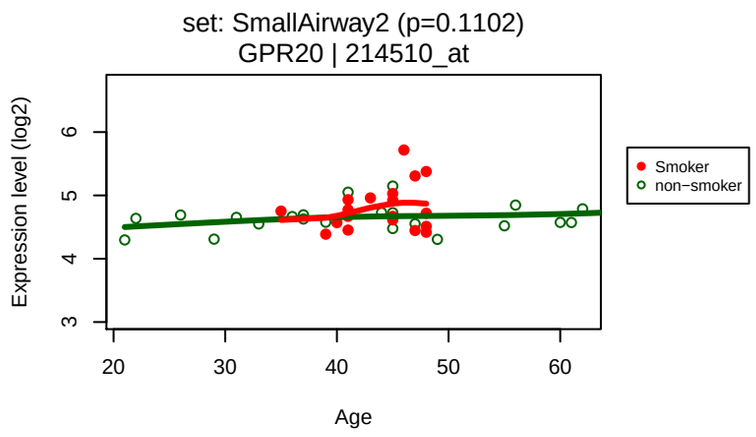
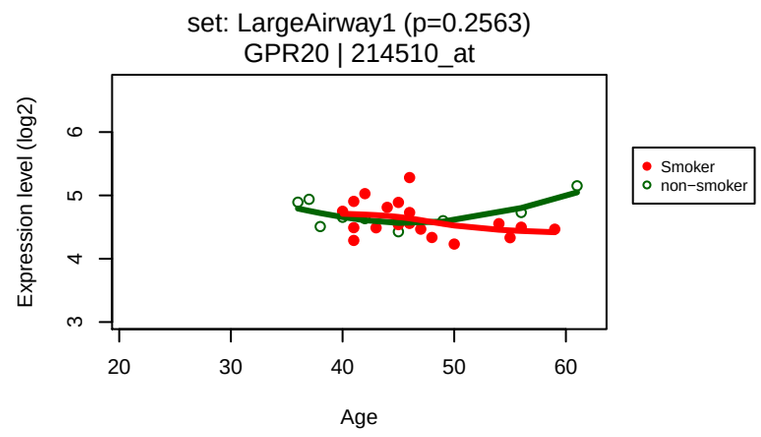

Gene expression of smokers and nonsmokers
gene: GPR20 | 214510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.101
p=0.1102
p=0.3443
p=0.2563
p=0.7175
p=0.2663
p=0.6507
p=0.5073
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

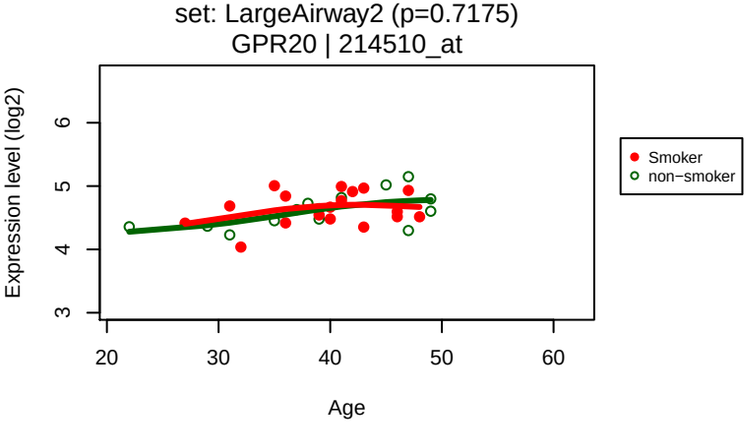

Gene expression ratio (smokers/nonsmokers)
gene: GPR20 | 214510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.101
p=0.1102
p=0.3443
p=0.2563
p=0.7175
p=0.2663
p=0.6507
p=0.5073
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

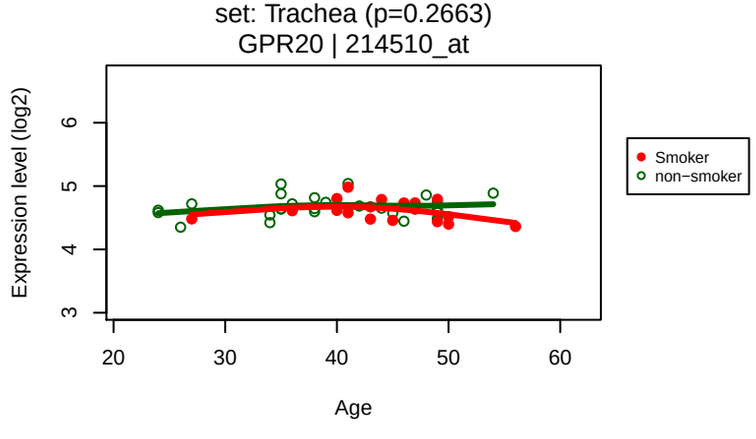

Gene expression ratio in Males
gene: GPR20 | 214510_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6292
p=0.6002
p=0.7069
p=0.2154
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR20 | 214510_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1829
p=0.659
p=0.0676
p=0.6499
SmallAirway2
LargeAirway2
Trachea
Macrophage2

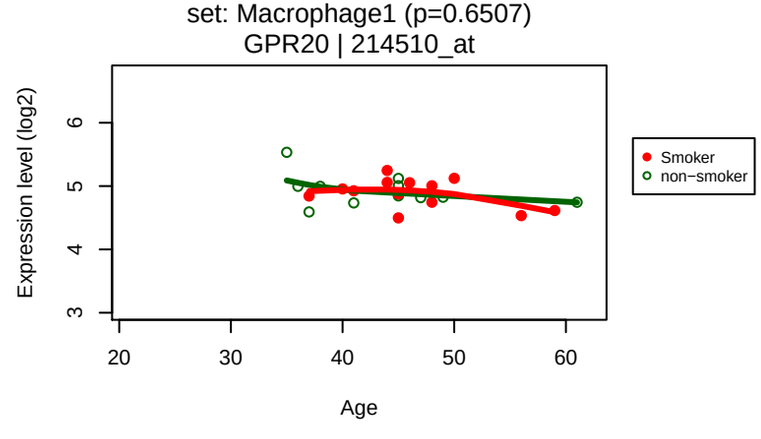

set: SmallAirway1 (p=0.101)
GPR20 | 214510_at
Expression level (log2)
Age
Smoker
non−smoker

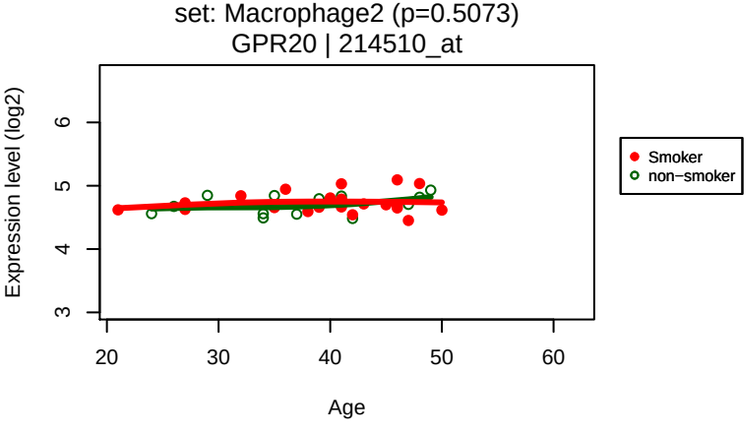

set: SmallAirway2 (p=0.1102)
GPR20 | 214510_at
Expression level (log2)
Age
Smoker
non−smoker

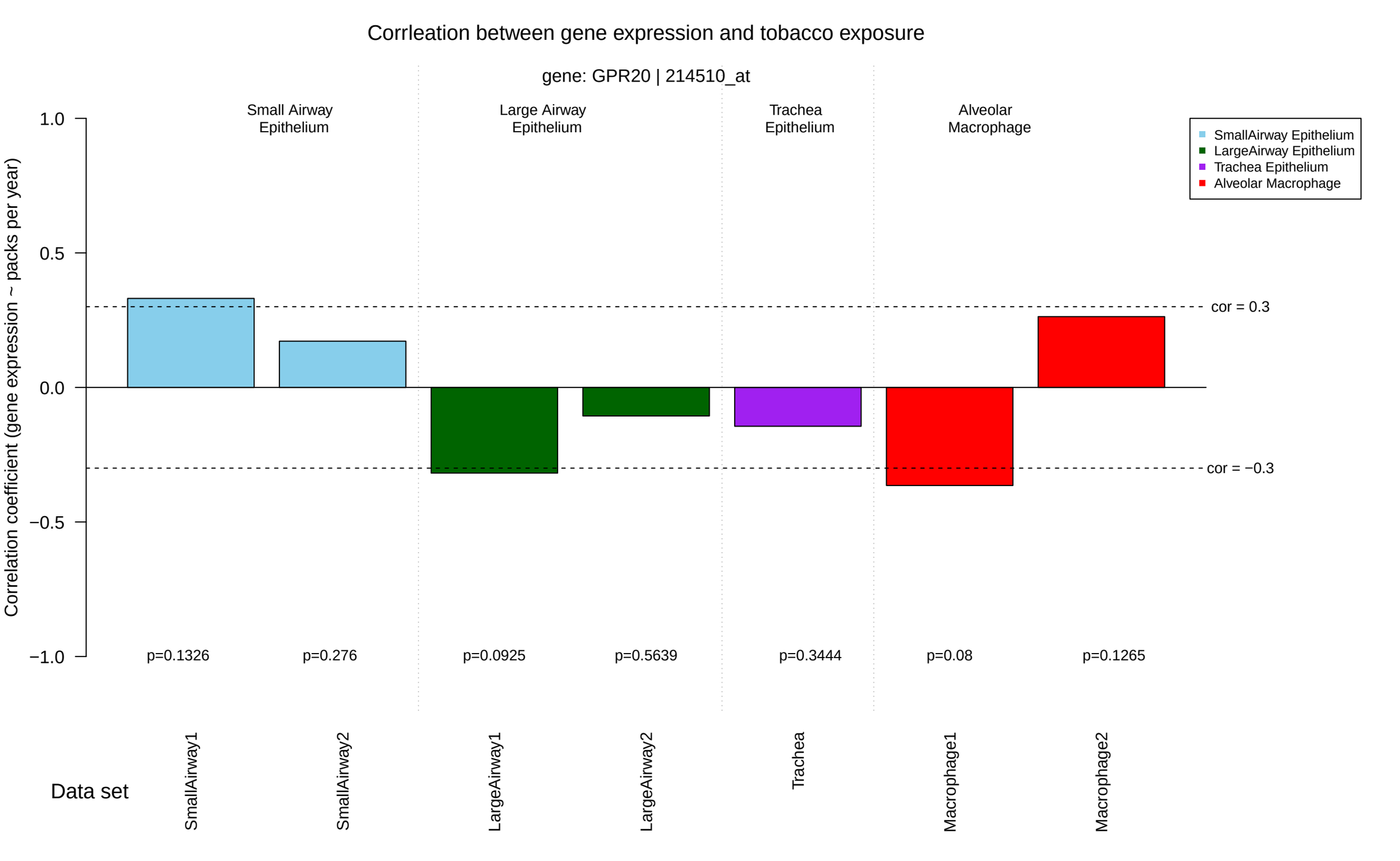

set: LargeAirway1 (p=0.2563)
GPR20 | 214510_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7175)
GPR20 | 214510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.2663)
GPR20 | 214510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6507)
GPR20 | 214510_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5073)
GPR20 | 214510_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR20 | 214510_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1326
p=0.276
p=0.0925
p=0.5639
p=0.3444
p=0.08
p=0.1265
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2